PAP Many Daughters project

Joop Adema

May 30, 2025

1 Introduction and design

This PAP specifies how we aim to test the four hypotheses stated in the Many Daughters project. Although getting a daughter is exogenous, it is correlated to various things. First of all, it is correlated to the number of children, which also correlates to age and time. Hence, we opt to always control for the number of children FE and age*sex (of the respondent) fixed effects and time fixed effects to absorb trends. Although the SOEP is longitudinal and has a fair amount of observations per person, some of the variables relevant to execute the project have very limited temporal coverage. Hence we rely on between individual variation, rather than relying on an individual FE design.

2 Variables and Data

In this section the item in the list corresponds to the variable name of the recoded variable and variable names from SOEP in TEXTSC.

2.1 Outcomes

- women_career
  In the 2016-2018 the SOEP included a battery of 7-point scale variables measuring attitudes towards gender equality PLJ0621-PLJ0625. However, this is only asked to those born abroad, which is a population incomparable to the general German population. Instead the 2012 wave included a question on PLH0301, which asks whether respondents think that woman should care about family rather than career (1 = fully agree, 4 = fully disagree). We remove all missings and no answers and keep the scale and interpret a higher score as preferences for gender equality. Although it only concerns one wave, it has been asked to a large part of the sample in 2012.

- leftleaningness:
  We use the variable on the left-right scale PLH0004. We remove missings and we invert the scale so higher=more left leaning.

- hours_chores:
  We use the differential number of hours spent on chores by men versus by women as a measure of gender equality. We use the variable PLH0301 and remove the missings.

2.2 Treatment

- d_daughter
  We interpret "Having daughters" as a binary indicator for having at least one daughter. We rely on the variables concerning children KIDSEX01-KIDSEX19 and KIDGEB01-KIDGEB19. We count the number of daughters by summing (over 01-19) whether KIDSEX=2 and kidgeb<syear (so the kid is born before the current year).

- male
  takes value 1 if SEX=1, 0 otherwise

2.3 Controls

- nkids:
  We rely on the variables concerning children KIDSEX01-KIDSEX19 and KIDGEB01-KIDGEB19. We count the number of children by summing (over 01-19) whether KIDSEX=1 OR 2 and kidgeb¡syear (so the kid is born before the current year).

- nkids_ever:
  We rely on the variables concerning children KIDSEX01-KIDSEX19 and KIDGEB01-KIDGEB19. We count the number of children by summing (over 01-19) whether KIDSEX=1 OR 2.

- age:
  We operationalize age as: SYEAR-GEBJAHR

- sex:
  This is simply SEX

- year:
  This is simply SYEAR

- partner_dum:
  We recode PARTNER as 1 if the person has a spouse, partner, or probably so (1-4) and 0 is someone does not have a partner (value=0), and missing if the partner variable is missing or if it is not clear (value=5).

- corigin_nm:
  We recode CORIGIN and remove the missings to get an indicator variable for country of birth.

- edu_level:
  We recode PGISCED97 and remove the missings to get an indicator variable for then level of educational attainment of an individual.

3 Inclusion/exclusion

In all models, we exclude individuals that never have any children in the sample period as those may be fundamentally different. We also include individuals aged below 18. We also implicitly exclude missing observations in the variables above. We focus on the full time period, expect for when variable availability limits this. A limitation of these restrictions is that they do not always pertain to the same sample because of limited coverage of the different outcome variables. However, as some variables have very limited coverage, we opt not to restrict the other outcome variables.

For H3 we explicitly restrict the sample to those individuals that have a partner at the time of interview.

4 Estimation

We estimate the following four equations and assess H1-H4 using $\beta_1 - \beta_4$:

$$genderequality_{it} = \beta_1 d_daughter + FE + \epsilon_{it} \quad (1)$$

$$leftleaningness_{it} = \beta_2 d_daughter + FE + \epsilon_{it} \quad (2)$$

$$hours_choresit = \alpha_3 d_daughter + \beta_3 d_daughter \times male + FE + \epsilon_{it} \quad (3)$$

$$genderequality_{it} = \alpha_4 d_daughter + \beta_4 d_daughter \times male + FE + \epsilon_{it} \quad (4)$$

For individual i interviewed at time t. The fixed effects are always interacted with sex, as the effects of specific factors (e.g. education) on gender equality may markedly differ for men and women. I include the following FE: age $\times$ sex, interview year $\times$ sex, number of children $\times$ sex, educational attainment $\times$ sex, and country of origin $\times$ sex.

In equation (3) we limit the sample for which partner_dummy is equal to one, as we otherwise would take into single-households where there is no notion of gender equality.

In all regressions, we cluster my standard errors on the (current) household level (HID), to account for intra-household correlation.

5 Hypotheses

We test the 4 hypotheses by testing whether the respective β's are positive and statistically significant at a 5% level.